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Evaluation of Segment Anything Model for Riparian Land Cover Classification from Aerial Imagery

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Abstract. Riparian land cover classification (LCC) is an important task for environmental monitoring and management, as it provides information on the state and characteristics of riparian zones, which are the transitional areas between terrestrial and aquatic ecosystems. However, LCC is challenging due to the high diversity and complexity of riparian vegetation and land use. In this paper, we evaluate the performance of Segment Anything Model (SAM), a recently proposed deep learning model for general image segmentation, for riparian LCC from aerial imagery. SAM can produce high quality object masks from input prompts such as points or boxes, and can generate masks for all objects in an image without fine-tuning. The authors compared SAM with the results derived from the DeepLabV3+ model trained by the dataset collected in Japan. Our experiments showed that SAM achieved competitive or superior results compared to other models, and can segment various riparian land cover types with high accuracy and consistency. The authors concluded that SAM is a promising tool for riparian LCC from aerial imagery, and suggest some directions for future research combining with the other model.

Keywords: DeepLabV3+, Land Cover Classification, Segment Anything Model, UAV

1 Introduction

Recent years, imagery derived from unmanned aerial vehicle (UAV) become an important tool for environmental monitoring. LCC mapping based on the UAV imagery is an efficient and critical approach for environmental researchers to collect the information on the specific location, especially the riparian area that are difficult for on-site survey ^[1]. In this context, the authors evaluated SAM-assisted LCC (i.e., LCC mapping preprocessed by Segment Anything Model ^[2] and classification judged by the researchers) and DeepLabV3+ ^[1] LCC Results derived from the same study site.

In the previous work ^[1], for understanding the land cover and land use with more detailed information, the authors divided the land cover classification into multiple labels (i.e., Clutter, Road, Grass and Bare Ground in Low-Vegetation/Ground; Bamboo and Tree in High-Vegetation). But if the LCC mapping was made under the mentioned classification standard, the Salt and Pepper noise (i.e., a kind of noise widely appeared in the segmentation tasks) always existed. At the same time, in the 2D (i.e., two-dimensional) flood simulation around water level change, the distribution of the high and low vegetation is most important fact that affect the results, the classification with detailed labels did not affect too much on the water level change.

Based on the mentioned reasons, and considering of the practical applications like riparian environment monitoring, especially between high and low vegetation, the authors separated the land cover classification into three main species: High-Vegetation, Low-Vegetation/Ground and Water.

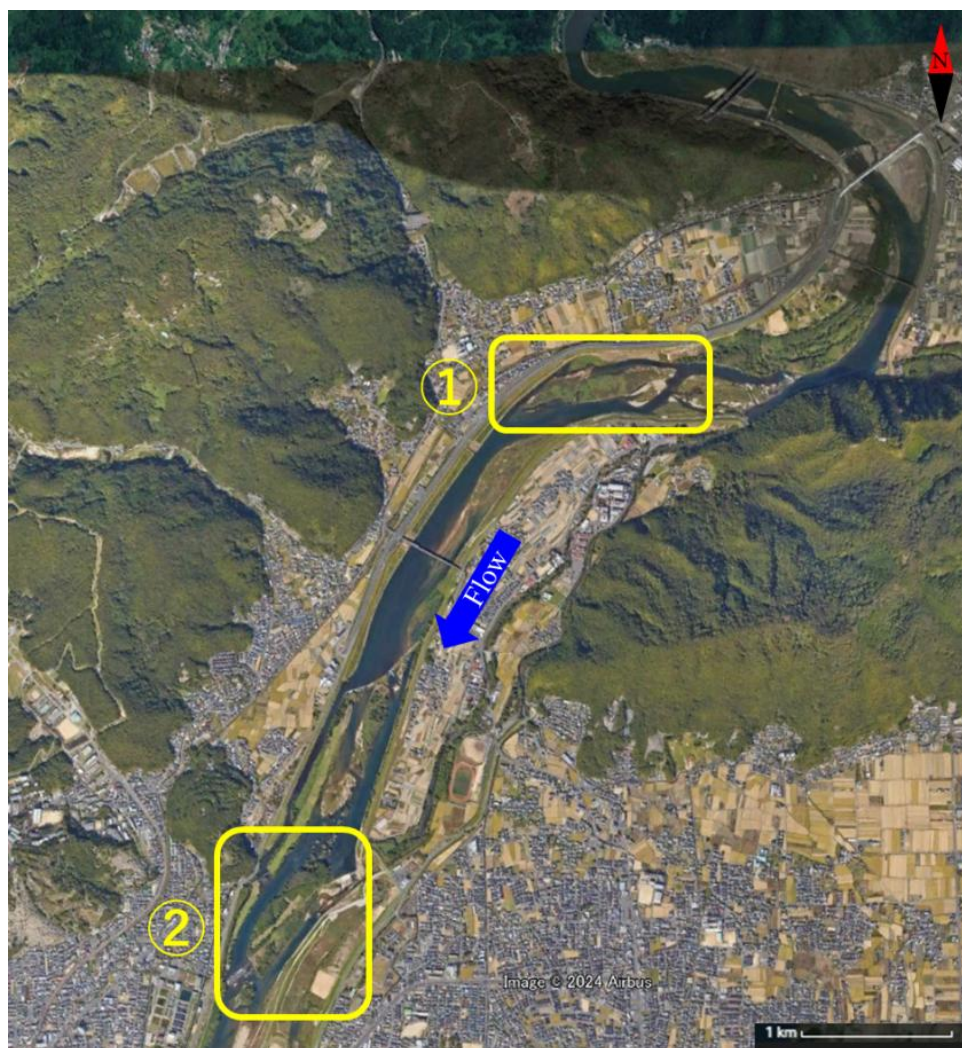


Figure 1 Two Study sites were selected in the Asahi River, Japan. Section-1 and -2 are namely Gion and Bunnryubu (i.e., the diversion weir of the Asahi River), respectively.

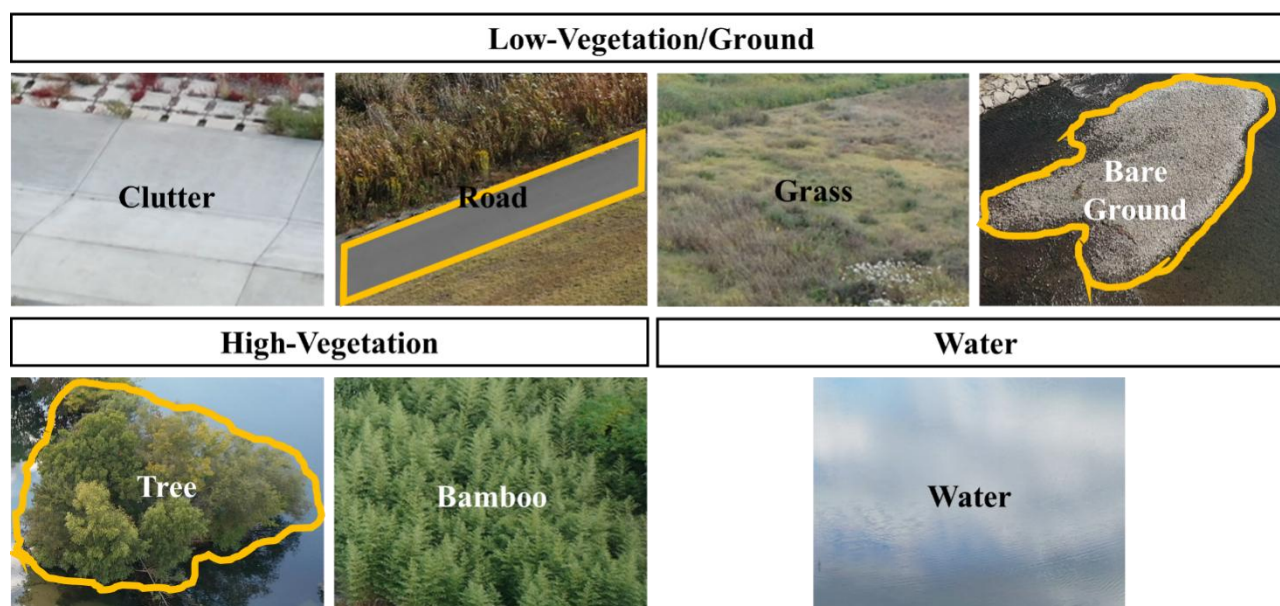


Figure 2 Sample images of Low-Vegetation/Ground, High-Vegetation and Water from the low altitude drone view.

2 Study site

Figure 1 shows the study sites in this paper, which are two downstream sections (i.e., section-1, Gion, and section-2, Bunnryubu) of the Asahi River, a first-class (state-controlled) river in Japan, which flows through Okayama Prefecture to the Seto Inland Sea. Figure 2 depicts the sample images of land cover in the study sites.

The reason for selecting these two sections is mainly based on the vegetation grown in the middle of the river channel. Because in the flood season, the river-channel-located high vegetation has the risk of raising the water level that affect the safety of the citizens living near the river, the deforestation is necessary depending on the previous flood situation. Before the deforestation, the work load assessment need to be considered. If the assessment on these sites with the high vegetation are accurate enough, it is available for assisting the policy makers to make the future decision.

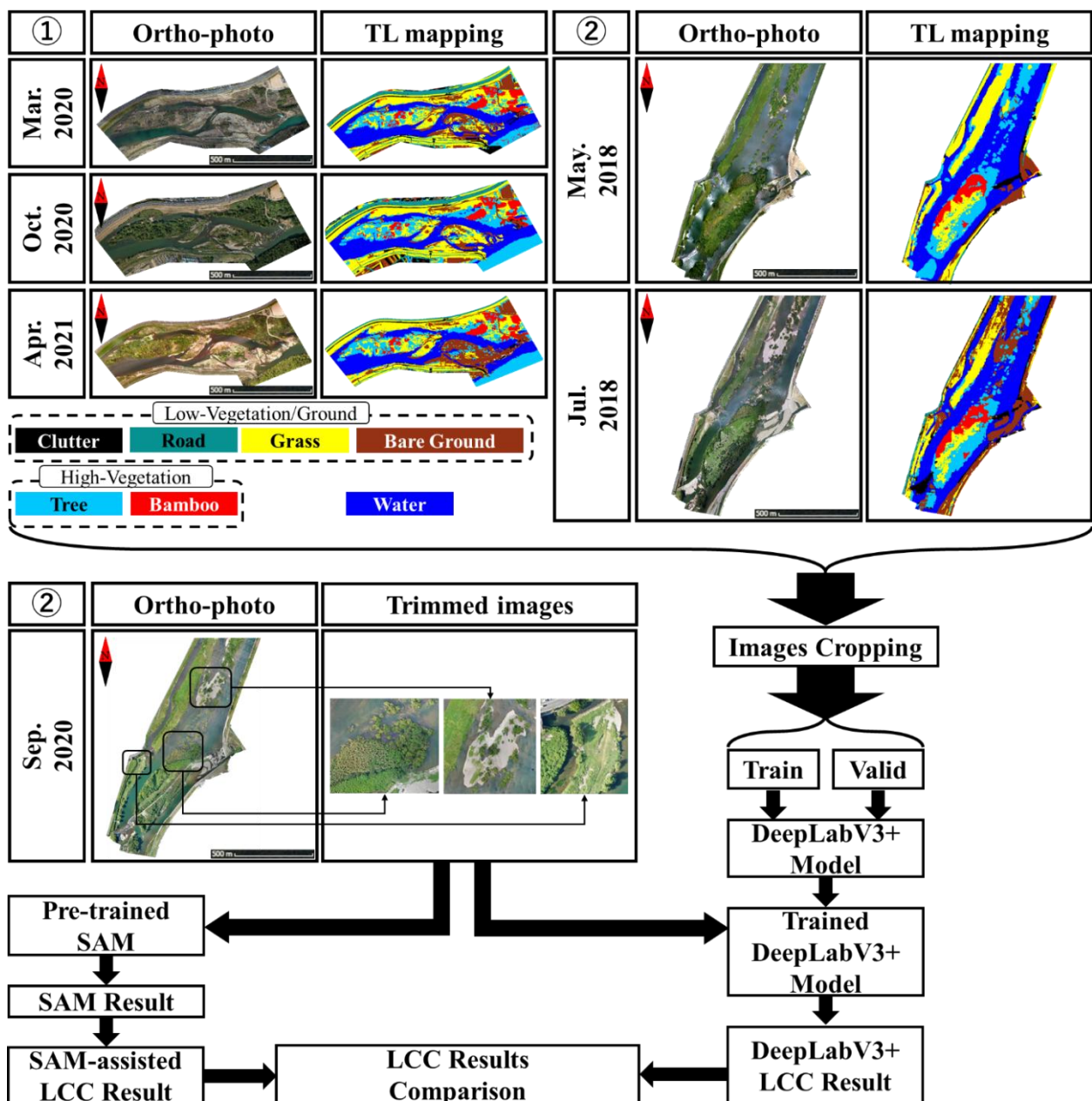


Figure 3 DeepLabV3+ and SAM-assisted LCC Results-based Comparison with Ortho-photo and TL of 3 locations.

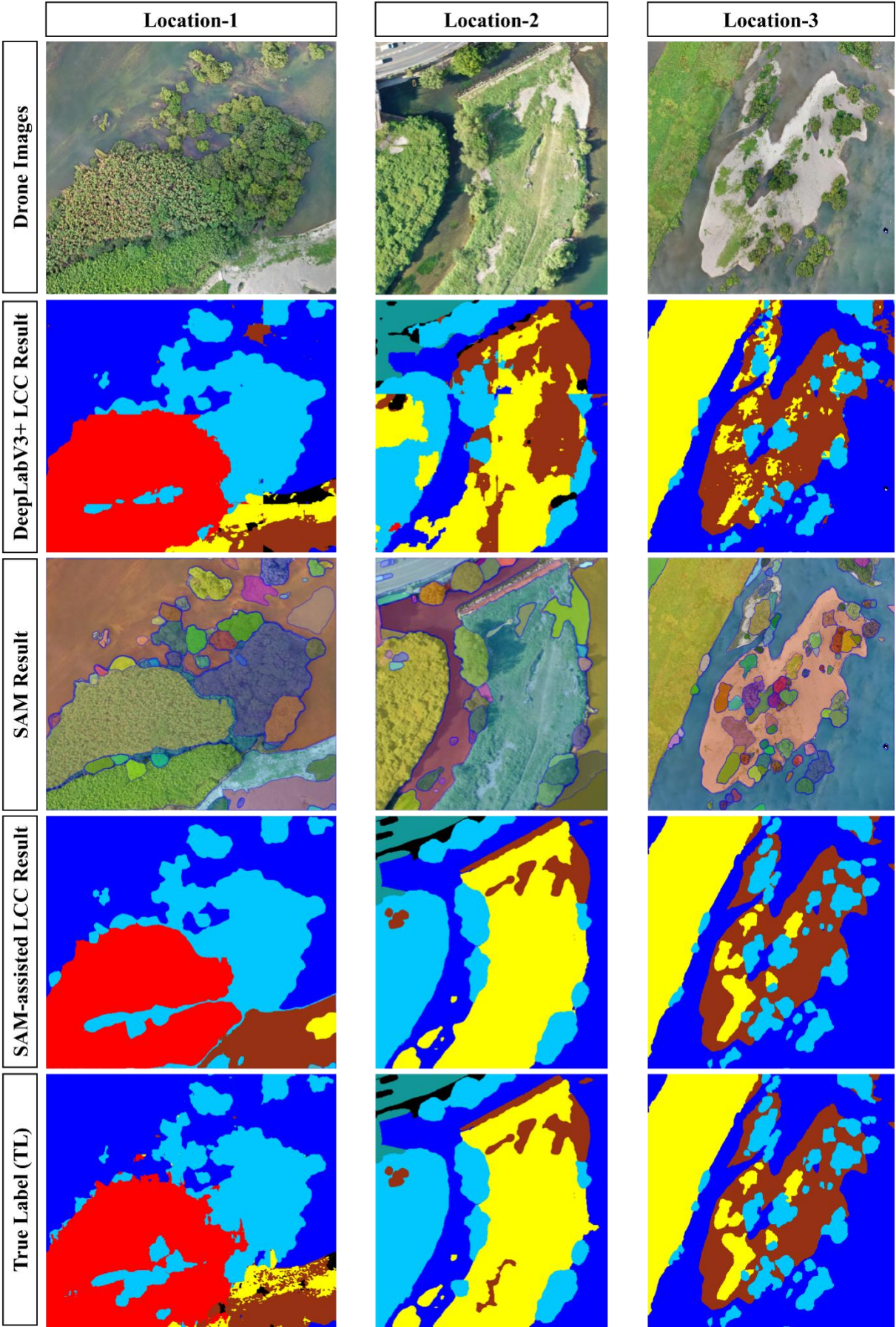
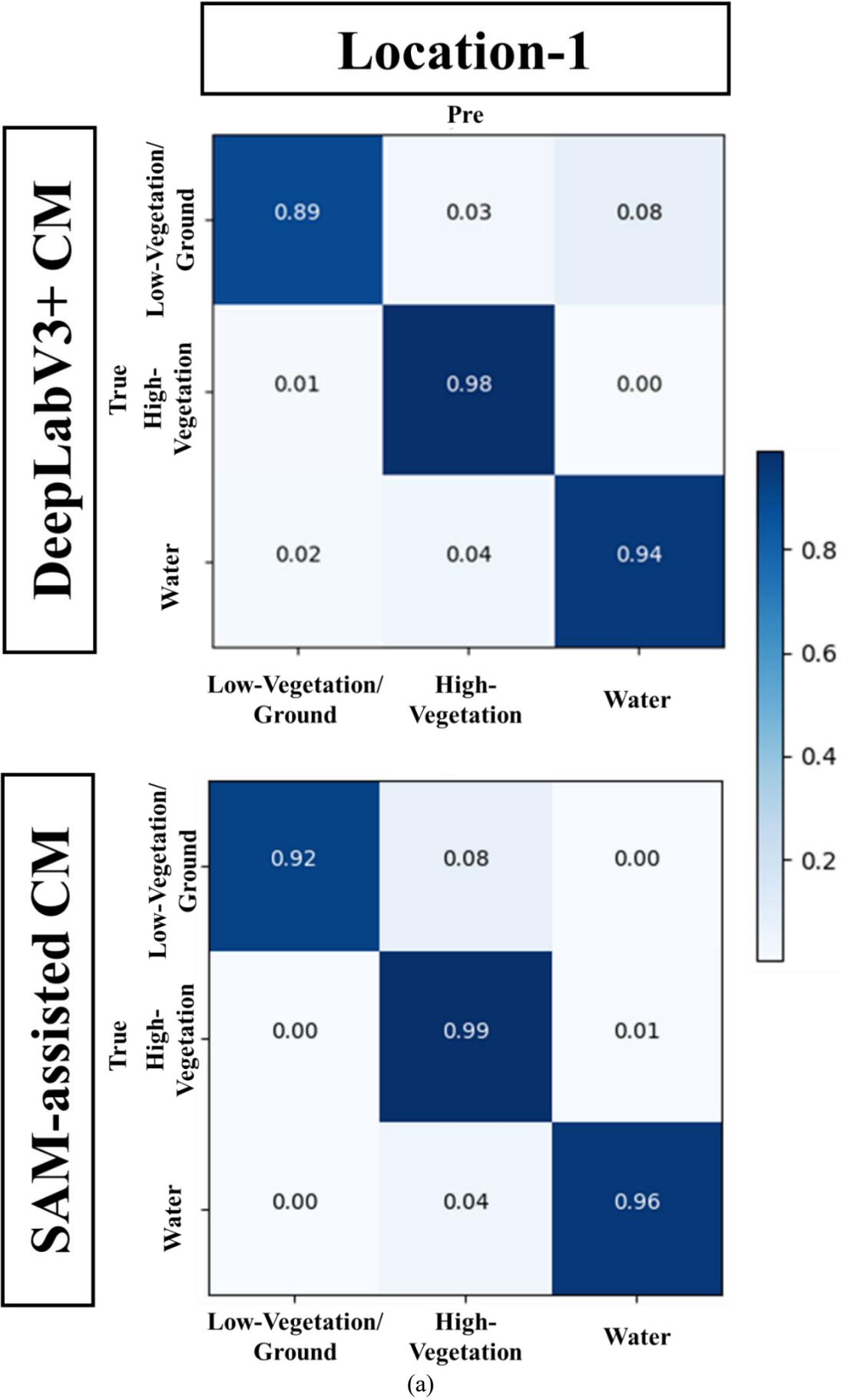
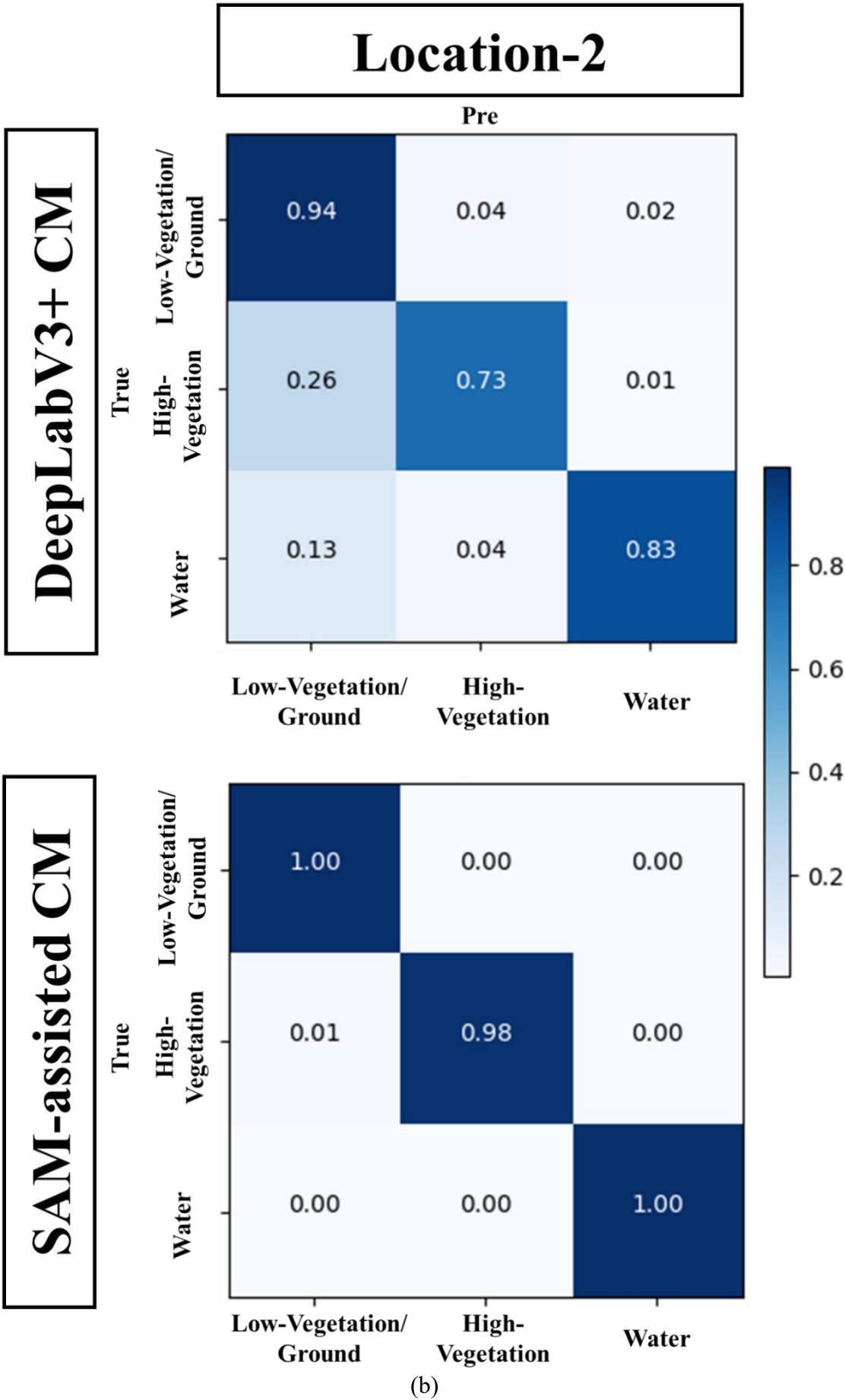


Figure 4 DeepLabV3+ and SAM-assisted LCC Results-based Comparison with Ortho-photo and TL of 3 locations.





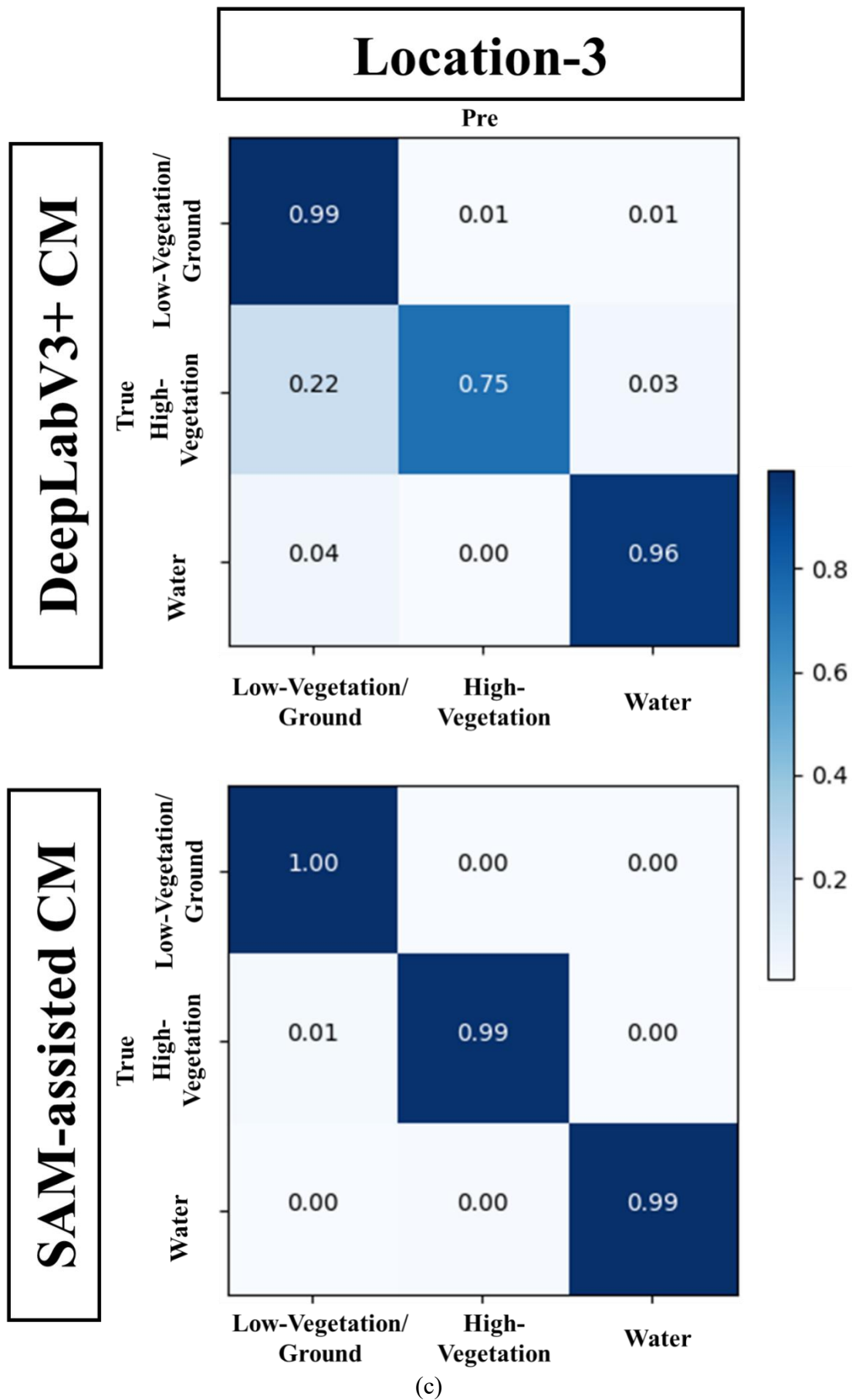


Figure 5 DeepLabV3+ and SAM-assisted LCC Confusion Matrix (CM) of 3 locations.

3 Work flow

In Figure 3, the work flow of this research is mainly divided into three parts: DeepLabV3+ LCC process, SAM-assisted LCC process and LCC Results Comparison. In the DeepLabV3+ LCC process, the authors need to train the custom dataset, and classifying the land cover without the artificial classification judgment. In the SAM-assisted LCC process, the artificial classification judgment need to be done after the SAM-based LCC mapping preprocessed. And based on the same trimmed images from 3 locations selected in Sep. 2020 Ortho-photo, the authors depicted the comparison between DeepLabV3+ and SAM-assisted LCC results.

In the DeepLabV3+ LCC process, the authors trained the DeepLabV3+ model using land cover dataset (train/valid) that include cropped ortho-photo and responding masks. There are several seasons of the ortho-photo (i.e., March, April, May, July and October) that were included in this dataset for training the DeepLabV3+ Model, increasing the robustness of the Trained DeepLabV3+ Model. As the final step of the DeepLabV3+ LCC process, the Trained DeepLabV3+ Model inferred the trimmed images as DeepLabV3+ LCC Results.

In the next step, the pre-trained SAM segmented the trimmed images from the same mentioned locations into individual patterns, namely SAM Results. Derived from the SAM Results, the authors re-classified the land cover, namely SAM-assisted LCC Results.

As the final step shown in the Figure 3, the authors compared the LCC Results derived from DeepLabV3+ and SAM-assisted with TL, respectively. The Figure 4 displayed the LCC mapping derived from individual methods (i.e., DeepLabV3+ LCC Results and SAM-assisted LCC Results) in 3 different locations. All of the mentioned locations for inferring include High-Vegetation, Low Vegetation/Ground and Water. The mentioned results-related comparisons were shown in the Figure 5 using CM (i.e., confusion matrix, a specific table for evaluating the accuracy of the machine learning results).

4 Results

As depicted in the Figure 5, the findings of this research reveal that the performance shown in CM (i.e., High-Vegetation, Low-Vegetation/Ground and Water) of SAM-assisted LCC Results is generally better than DeepLabV3+ LCC Results. Especially the High-Vegetation and Low-Vegetation/Ground were misclassified with each other using the DeepLabV3+ Model. Using both methods, the water can be classified with over 0.9 Recall value, just except of the location-2 using DeepLabV3+ Model.

5 Conclusion

In this study, SAM-assisted LCC Results is better than the LCC Results derived from the DeepLabV3+. But the artificial judgment is also in need in the SAM-assisted LCC Results. The collaboration of these mentioned methods is in the consideration for the future research.

Reference

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